

Table S1(b): Model GWL Crossing Years and Model Group Assignments (SSP2-4.5)

Model	Scenario	GWL +2C Year	Winter Model Group (+2C)	Summer Model Group (+2C)
ACCESS-CM2	SSP2-4.5	2041	Low-Frequency	High-Frequency
ACCESS-ESM1-5	SSP2-4.5	2045	High-Frequency	High-Frequency
CESM2	SSP2-4.5	2046	High-Frequency	High-Frequency
CESM2-WACCM	SSP2-4.5	2039	High-Frequency	High-Frequency
CMCC-CM2-SR5	SSP2-4.5	2038	Low-Frequency	High-Frequency
CMCC-ESM2	SSP2-4.5	2041	High-Frequency	Low-Frequency
CanESM5	SSP2-4.5	2024	Low-Frequency	Low-Frequency
EC-Earth3	SSP2-4.5	2044	High-Frequency	Low-Frequency
EC-Earth3-CC	SSP2-4.5	2035	Low-Frequency	Neutral / Other
EC-Earth3-Veg-LR	SSP2-4.5	2050	Low-Frequency	Low-Frequency
GFDL-ESM4	SSP2-4.5	2073	Low-Frequency	Low-Frequency
HadGEM3-GC31-LL	SSP2-4.5	2034	High-Frequency	High-Frequency
IITM-ESM	SSP2-4.5	2064	Low-Frequency	Low-Frequency
INM-CM4-8	SSP2-4.5	2064	High-Frequency	High-Frequency
INM-CM5-0	SSP2-4.5	2072	High-Frequency	Low-Frequency
IPSL-CM6A-LR	SSP2-4.5	2034	Neutral / Other	Low-Frequency
KACE-1-0-G	SSP2-4.5	2024	High-Frequency	High-Frequency
KIOST-ESM	SSP2-4.5	2042	Low-Frequency	High-Frequency
MIROC-ES2L	SSP2-4.5	2064	Low-Frequency	Low-Frequency
MIROC6	SSP2-4.5	2075	High-Frequency	Low-Frequency
MPI-ESM1-2-HR	SSP2-4.5	2061	High-Frequency	High-Frequency
MPI-ESM1-2-LR	SSP2-4.5	2060	Low-Frequency	Low-Frequency
MRI-ESM2-0	SSP2-4.5	2049	Low-Frequency	Low-Frequency
NorESM2-MM	SSP2-4.5	2081	High-Frequency	High-Frequency
UKESM1-0-LL	SSP2-4.5	2035	Low-Frequency	High-Frequency